

# SustainX EI99 Report

**Model:** Lower Casing Trial

**Type:** Part

**Total EI99:** 27,598.56 mPt

## Production

| Item                            | Basis      | Indicator     | Points (mPt) |
|---------------------------------|------------|---------------|--------------|
| Material production (Cast iron) | 134.358 kg | 240.00 mPt/kg | 32,245.84    |

## Processing

| Item                        | Basis      | Indicator      | Points (mPt) |
|-----------------------------|------------|----------------|--------------|
| Drilling (holes)            | 0.1445 dm3 | 800.00 mPt/dm3 | 115.61       |
| Milling (area x 1mm)        | 0.0637 dm3 | 800.00 mPt/dm3 | 50.97        |
| Boring (area x 1mm)         | 0.0658 dm3 | 800.00 mPt/dm3 | 52.62        |
| Grinding (area x 1mm)       | 0.1653 dm3 | 800.00 mPt/dm3 | 132.21       |
| Casting (fuel: Natural Gas) | 483.32 MJ  | 5.30 mPt/MJ    | 2,561.62     |

## Recycling

| Item                               | Basis      | Indicator     | Points (mPt) |
|------------------------------------|------------|---------------|--------------|
| Recycling (Recycling Ferro metals) | 133.317 kg | -70.00 mPt/kg | -9,332.20    |

## Transport

| Item              | Basis                | Indicator    | Points (mPt) |
|-------------------|----------------------|--------------|--------------|
| Freighter oceanic | 0.1344 t x 11,989 km | 1.10 mpt/tkm | 1,771.90     |

## Simulation Summary

**Iterations:** 8

**Final FoS Min:** 5.779

**Final Max Stress:** 31.89 MPa

**Pareto best iteration:** 5 (EI99=28,382.10 mPt, FoS min=6.241)

### Iterations

| Iteration | THICKNESS@Flange (mm) | THICKNESS@Bearing Sketch (mm) | Mass (kg) | Max Stress (MPa) | FoS Min | EI99 (mPt) |
|-----------|-----------------------|-------------------------------|-----------|------------------|---------|------------|
| 1         | 30.000                | 20.000                        | 141.781   | 22.50            | 6.210   | 29,433.98  |
| 2         | 29.000                | 19.000                        | 140.562   | 23.46            | 6.376   | 29,169.94  |

|   |        |        |         |       |       |           |
|---|--------|--------|---------|-------|-------|-----------|
| 3 | 28.000 | 18.000 | 139.345 | 24.66 | 6.391 | 28,906.62 |
| 4 | 27.000 | 17.000 | 138.133 | 26.56 | 6.184 | 28,644.00 |
| 5 | 26.000 | 16.000 | 136.924 | 27.73 | 6.241 | 28,382.10 |
| 6 | 25.000 | 15.000 | 135.718 | 29.37 | 5.778 | 28,120.91 |
| 7 | 24.000 | 14.000 | 134.516 | 31.44 | 5.546 | 27,860.43 |
| 8 | 23.000 | 13.000 | 133.317 | 31.89 | 5.779 | 27,600.65 |

## Trade-off Plot

